

Quiz Binary Numbers

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- ① Represent 00101100 in decimal & hexadecimal!

Decimal = 44

$$\begin{array}{cccccccc} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 128 & 64 & 32 & 16 & 8 & 4 & 2 & 1 \end{array} = 32 + 8 + 4 = 44$$

Hexadecimal = 2C

$$\begin{array}{cccc|cccc} 0 & 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 8 & 4 & 2 & 1 & 8 & 4 & 2 & 1 \end{array}$$

2 8+4 = 12
12 = C

- ② Represent decimal 123 in binary!

$$\begin{array}{ll} 123 \div 2 & 61 \dots 1 \\ 61 \div 2 & 30 \dots 1 \\ 30 \div 2 & 15 \dots 0 \\ 15 \div 2 & 7 \dots 1 \\ 7 \div 2 & 3 \dots 1 \\ 3 \div 2 & 1 \dots 1 \\ 1 \div 2 & 0 \dots 1 \end{array} \rightarrow 1111011$$

- ③ Calculate the result of adding 11011011 and 0110010!

$$\begin{array}{rcl} 1111 & & \\ 11011011 & = & 219_{(10)} \\ 01110010 & + & 114_{(10)} \\ \hline 101001101 & = & 333_{(10)} \rightarrow \text{overflow} \\ 01001101 & = & 77_{(10)} \end{array}$$

$333_{(10)} \bmod 2^8 = 77_{(10)}$